MILITARY SPECIFICATION

BAROSWITCH ASSEMBLY, M5

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of switch entitled M5 Baroswitch Assembly (BSA) which is used in the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-B-117	- Bags, Interior Packaging.
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.
MIL-P-17667	- Paper, Wrapping, Chemically Neutral.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For.

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DRAWINGS

ORDNANCE CORPS

8786179	- Terminal and Plate Assembly.
8786200	- M5 Baroswitch Assembly.
ET-8786200	- Equipment Tabulation.
8803825	- Plate and Terminal Assembly.
8803840	- Element Right Hind Assembly (S2).
88003842	- Element Left Hind Assembly (S3).
8803843	- Element Right Hind Assembly (S1).
8803844	- Element Left Hind Assembly (S4).
8803846	- Installation Outline.
8803863	- Terminal Lug.
8803870	- Gear Train Assembly.
8803877	- Spring.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1	- Quality Assurance Instructions for Reporting Data on Serialized Items.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The M5 baroswitch assembly shall comply with all requirements specified on the drawing (dwg) 8786200, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

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a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Operating tolerances. Operating tolerance is specified as a maximum allowable deviation in contact closure altitude from that operating altitude indicated on a remote indicator interconnected to the altitude indicating transmitter system of the baroswitch assembly (bsa). See 6.7 for information on transmitter system. Each baroswitch element (bse) shall be capable of meeting the requirements of Table I. The operating point of each bsa shall be within the tolerances specified when tested as specified in 4.4.1.

Table I

BSA Nominal Altitude Setting	Nominal Pressure Millibars (mb)	Maximum Allowance Operating Tolerance (mb)	Repeatability (mb) (max.)
-1000	1050.4	± 6.6	3.9
1000	977.2	± 5.3	1.6
4500	859.0	± 5.3	1.6
8000	752.6	± 5.3	1.6
15,000	571.8	± 9.6	1.6
25,000	376.0	± 16.0	1.6

Test Codes for Contact Closure Altitudes
(see 3.14, and 6.1.2)

25,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	030001	030004	030007	0300010
41 - 46	2	030002	030005	030008	0300011
60 - 65	3	030003	030006	030009	0300012

15,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	030014	030017	030020	030023
41 - 46	2	030015	030018	030021	030024
60 - 65	3	030016	030019	030022	030025

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8,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	030026	030029	030032	030035
41 - 46	2	030027	030030	030033	030036
60 - 65	3	030028	030031	030034	030037

4,500 feet

Record Readings in Blocks	Run No.	Test Identification Code Number BSE No. 1	2	3	4
22 - 27	1	030038	030041	030044	030047
41 - 46	2	030039	030042	030045	030048
60 - 65	3	030040	030043	030046	030049

+1,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Number BSE No. 1	2	3	4
22 - 27	1	030050	030053	030056	030059
41 - 46	2	030051	030054	030057	030060
60 - 65	3	030052	030055	030058	030061

-1,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Number BSE No. 1	2	3	4
22 - 27	1	030062	030065	030068	030071
41 - 46	2	030063	030066	030069	030072
60 - 65	3	030064	030067	030070	030073

3.6 Repeatability. Within a 24-hour period for any constant temperature and constant linear acceleration, each pressure cycle must produce a contact closure within 1.6 mb. above zero feet setting and within 3.9 mb. below zero feet of all other contact closure altitudes indicated by that element within the above 24-hour period when tested as specified in 4.4.1.

3.7 Response. The operating error of each element assembly in each bsa shall not exceed +5.3 (-168 feet), -6.7 (+210 feet) mb when tested as specified in 4.4.2. At the conclusion of this test the bsa shall be capable of meeting the requirements of 3.5 and 3.6.

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3.8 Pneumatic seal. The baroswitch assembly shall be assembled in such a manner as to minimize leakage. A sealing test shall be performed at +80 ± 15°F. The pressure change rate within the pneumatic system and the interconnected internal volume of the test equipment shall not exceed 3.9 mb/minute for a total system volume of 10 cubic inches. For test systems of any other total volumes, the allowable pressure change rate shall be inversely proportional to the volume ratio; i.e., for a system of 20 cubic inches (twice the above total volume) the allowable leakage rate would be one-half or 1.95 mb/minute when tested as specified in 4.4.3. Test code 060000.

3.9 Setting mechanism.

3.9.1 Range setting limits. The bsa shall be capable of being remotely set to any altitude setting between 25,000 (+700 -0), feet to -3,100 (+0, -700) feet. The remote setting shall be accomplished by means of electrical connections only (see 6.5.3 and 6.6). The setting mechanism shall be automatically stopped between 25,100 and 25,800 feet at the upper limit and between -3.100 and -3,800 feet at the lower limit of the setting range when tested as specified in 4.4.4. Test codes: Upper limit 071002, Lower limit 071003.

3.9.2 Remote setting rate. The setter mechanism shall be capable of changing the contact operation point at a rate not less than 7,000 feet of altitude per minute when tested as specified in 4.4.5. Test code 072001.

3.9.3 Synchro calibration. The deviation of the fine synchro transmitter from electrical zero shall not exceed +1.5 degrees (+15) feet when tested as specified in 4.4.10. Test code 073000.

3.10 Resistance.

3.10.1 Continuity loop. The resistance between pins X and B of the connector shall not exceed 0.1 ohms when tested as specified in 4.4.6.1. Test code 081000.

3.10.2 Element contact loop. The resistance of each element contact circuit measured at the pins of the electrical receptacle shall not exceed 0.4 ohm at a pressure of 3.3 mb greater than the operating pressure. The resistance shall be measured with the contacts closed and conducting a direct current of 300 milliamperes (ma) at 30 volts max. direct current (dc) when tested as specified in 4.4.6.2. Test code 082000.

3.10.3 Synchro rotor winding. The dc resistance of the two synchro rotors in parallel shall not exceed 16.3 ohms measured at the pins of the connector when tested as specified in 4.4.6.3. Test code 083000.

3.11 Insulation resistance.

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3.11.1 Element circuits. The insulation resistance of each baroswitch element circuit shall exceed 25 megohms when 500 ± 50 volts dc is applied at an altitude setting of -1000 feet and shall exceed 15 megohms when 300 ± 30 volts dc is applied at an altitude setting of 25,000 feet when tested as specified in 4.4.7.1.

Test codes at low altitude

Switch Position C:	910001
Switch Position P:	910002
Switch Position V:	910003
Switch Position M:	910004
Switch Position C:	910005
Switch Position D:	910006
Switch Position P:	910007
Switch Position W:	910008
Switch Position V:	910009
Switch Position L:	910010
Switch Position M:	910011
Switch Position N:	910012

3.11.2 Synchro and motor circuits. The insulation resistance shall exceed 1.5 megohms when 375 ± 25 volts root mean square, (vrms), 60 Hertz, (Hz) alternating current (ac) is applied between the synchro and motor circuits and common ground when tested as specified in 4.4.7.2. Test code 092000.

3.12 Drift and leakage and test codes. The bsa shall function as required in 3.5 and 3.6 after being tested as specified in 4.4.8.

Test Codes:

Initial Operating Readings

Record Readings in Blocks	Run No.	Test Identification Code Number BSE No. 1	2	3	4
22 - 27	1	100001	100004	100007	100010
41 - 46	2	100002	100005	100008	100011
60 - 65	3	100003	100006	100009	100012

Second 15 Day Drift

Record Readings in Blocks	Run No.	Test Identification Code Number BSE No. 1	2	3	4
22 - 27	1	100013	100016	100019	100022
41 - 46	2	100014	100017	100020	100023
60 - 65	3	100015	100018	100021	100024

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Third 15 Day Drift

Record Readings in Blocks	Run No.	Test Identification Code Number BSE No. 1	2	3	4
22 - 27	1	100025	100028	100031	100034
41 - 46	2	100026	100029	100032	100035
60 - 65	3	100027	100030	100033	100036

Fourth 15 Day Drift

Record Readings in Blocks	Run No.	Test Identification Code Number BSE No. 1	2	3	4
22 - 27	1	100037	100040	100043	100046
41 - 46	2	100038	100041	100044	100047
60 - 65	3	100039	100042	100045	100048

3.13 Environmental testing. The bsa shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The bsa shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.9. Tests shall be conducted in the order specified.

3.13.1 Low temperature.

3.13.1.1 Remote setting rate. The setting mechanism shall be capable of changing the contact operating point at a rate not less than 7,000 feet of altitude per minute at a stabilized temperature (see 6.4) of -65°F. ± 5°F. when tested as specified in 4.4.9.1. Test code 011101.

3.13.1.2 Pneumatic seal. The baroswitch assembly shall be assembled in such a manner as to minimize leakage. A sealing test shall be performed with the pneumatic system subjected to a pressure of 666 ± 6.7 mb below the external pressure. At -65 ± 5°F., the pressure change rate within the pneumatic system and the interconnected internal volume of the test equipment shall not exceed 8.0 mb/minute for a total system volume of 10 cubic inches. For test systems of any other total volumes, the allowable pressure change rate shall be inversely proportional to the volume ratio; i.e., for a system of 20 cubic inches (twice the above total volume) the allowable leakage rate would be one-half or 4.0 mb/minute when tested as specified in 4.4.9.1.2. Test code 011102.

3.13.1.3 Operating tolerance and repeatability. The bsa shall meet the requirements of 3.5 and 3.6 at a stabilized temperature (see 6.4) of -65 ± 5°F. in lieu of the ambient temperatures when tested as specified in 4.4.9.1.3.

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Test Codes for Contact Closure Altitudes - (Low Temperature)

25,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Number BSE No. 1	2	3	4
22 - 27	1	111301	111304	111307	111310
41 - 46	2	111302	111305	111308	111311
60 - 65	3	111303	111306	111309	111312

15,000 feet

Record Reading in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	111313	111316	111319	111322
41 - 46	2	111314	111317	111320	111323
60 - 65	3	111315	111318	111321	111324

8,000 feet

Record Reading in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	111325	111328	111331	111334
41 - 46	2	111326	111329	111332	111335
60 - 65	3	111327	111330	111333	111336

4,500 feet

Record Reading in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	111337	111340	111343	111346
41 - 46	2	111338	111341	111343	111347
60 - 65	3	111339	111342	111344	111348

+1,000 feet

Record Reading in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	111349	111352	111355	111358
41 - 46	2	111350	111353	111356	111359
60 - 65	3	111351	111354	111357	111360

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-1,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	111361	111364	111367	111370
41 - 46	2	111362	111365	111368	111371
60 - 65	3	111363	111366	111369	111372

3.13.2 High temperature.

3.13.2.1 Remote setting rate. The bsa shall meet the requirements of 3.13.1.1 except that the temperature shall be stabilized at (see 6.4) +200 ± 5°F. in lieu of -65 ± 5°F. when tested as specified in 4.4.9.2.1. Test code 011200.

3.13.2.2 Operating tolerance and repeatability. The bsa shall meet the requirements of 3.5 and 3.6 at a stabilized temperature (see 6.4) of 200 ± 5°F. in lieu of ambient temperature when tested as specified in 4.4.9.2.2.

Test Codes for Contact Altitudes
(High Temperature)

25,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	112201	112204	112207	112210
41 - 46	2	112202	112205	112208	112211
60 - 65	3	112203	112206	112209	112212

15,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	112213	112216	112219	112222
41 - 46	2	112214	112217	112220	112223
60 - 65	3	112215	112218	112221	112224

8,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	112225	112228	112231	112234
41 - 46	2	112226	112229	112232	112235
60 - 65	3	112227	112230	112233	112236

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4,500 feet

Record Readings in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	112237	112240	112243	112246
41 - 46	2	112238	112241	112244	112247
60 - 65	3	112239	112242	112245	112248

+1,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	112249	112252	112255	112258
41 - 46	2	112250	112253	112256	112259
60 - 65	3	112251	112254	112257	112260

-1,000 feet

Record Readings in Blocks	Run No.	Test Identification Code Numbers BSE No. 1	2	3	4
22 - 27	1	112261	112264	112267	112270
41 - 46	2	112262	112265	112268	112271
60 - 65	3	112263	112266	112269	112272

3.13.3 Storage.

3.13.3.3 Low temperature. The bsa shall be capable of withstanding temperatures of -65 ± 5°F. and -80 ± 5°F. and the average operating point after the environment shall be within 1.6 mb. of the average operating point before the environment when tested as specified in 4.4.9.3.1. After this test the bsa shall be capable of meeting the requirements of 3.5, 3.6, 3.8 through 3.12 except that the operating tolerance may be recalibrated, if necessary, to assure optimum operation within the required operating tolerances of 3.5 and 3.6.

3.13.3.2 High temperature. The bsa shall be capable of withstanding temperatures from +90 ± 5°F. to +160 ± 5°F. and the average operating point after the environment shall be within 1.6 mb of the average operating point after the environment when tested as specified in 4.4.9.3.2. After this test the bsa shall then be capable of meeting the requirements of 3.5, 3.6, and 3.8 through 3.11 except that the operating tolerance may be increased 1.6 mb. If the bsa meets these test requirements it may be recalibrated, if necessary, before shipment, to assure optimum operation within the required operating tolerances of 3.5 and 3.6.

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3.13.4 Temperature shock. The bsa shall be capable of withstanding three cycles at +160 ± 5°F. for 4 hours and -65 ± 5°F. for 4 hours, and within 1 hour of the last test cycle shall meet the requirements of 3.5, 3.6, and 3.8 through 3.11 when tested as specified in 4.4.9.4.

3.13.5 Vibration.

3.13.5.1 Contact chatter. The bsa shall be capable of withstanding vibration in each of three mutually perpendicular axis with no indication of contact chatter when tested as specified in 4.4.9.5 and 4.4.9.5.1. At the conclusion of this test the bsa shall be capable of meeting the requirements of 3.5, 3.6, 3.8, 3.9.1, 3.10 and 3.11.

3.13.5.2 Resonance. The bsa shall be capable of meeting the requirements of 3.4, 3.7 and 3.9 through 3.12 after being tested as specified in 4.4.9.5.2.

3.13.6 Acceleration. The bsa shall be capable of withstanding a 50 gravity (g) acceleration in any direction without effect on operation of assembly. The switch elements shall be capable of operating within ± 2.0 mb's of the static operating point when exposed to 15 g maximum acceleration in any direction when tested as specified 4.4.9.6.

3.13.7 Shock. The bsa shall be capable of operating as specified in 3.5 through 3.7 and shall have no visible damage after being subjected to impact shocks, 12 of 30 g and six of 50 g. Each 30 g shock impulse shall reach its maximum within 0.0055 ± 0.001 second and shall have a total duration of 0.011 ± 0.001 second; each 50 g shock impulse shall reach its maximum within 0.0025 second and shall have a total duration of 0.005 ± 0.001 second when tested as specified in 4.4.9.7.

3.13.8 Overpressure. The final average operating pressure of each bse shall be within 1.6 mb of the initial reading after the bsa has been subjected to pressure of 1520 ± 5 mb absolute for 20 hours when tested as specified in 4.4.9.8.

3.13.9 Under pressure. The final average operating pressure of each bse shall be within 1.6 mb of initial reading after the bsa has been subjected to pressure of 10.6 mb absolute for 20 hours when tested as specified in 4.4.9.9.

3.13.10 Expected life. The remote setting mechanism of the bsa assembly shall be capable of 300 adjusting cycles. One adjusting cycle is defined as the adjustment from -3000 to 25,000 and back to -3000 feet within the allowable setting range and tolerances of 3.11 when tested as specified in 4.4.9.10. The bsa shall meet the requirements of 3.5, 3.6, 3.8 through 3.11 except that loop resistance limit of 3.10.2 shall be increased to 0.5 ohms and operating error tolerances shall be increased 2.0 mb. If after the test the bsa has been found acceptable it may be recalibrated to assure optimum operation within the requirements of 3.5 and 3.6.

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3.13.11 Pressure and contact cycling. The bsa pressure cavity shall be capable of being subjected to 1000 cycles of pressure from 1014 to 10.7 to 1014 mb and during the final 500 pressure cycles the bsa contacts shall make and break a 100 + 10 ma current in a non-inductive circuit of 30 + 5 volts dc when tested as specified in 4.4.9.11. At the conclusion of this test, the bsa will meet the requirements of 3.5, 3.6, 3.8, 3.10 and 3.11 except that the loop resistance limit noted in 3.10.2 shall be increased to 0.5 ohms.

3.14 Test codes. The contractor shall provide data feed back on punch cards in conformance with Publication QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraphs in this section.

3.15 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.16 Workmanship.

3.16.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.16.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.16.3 Burrs. Parts shall be free from burrs.

3.16.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.16.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The contractor's quality assurance system shall contain a procedure for the reporting of critical defects found during the course of his inspection and during inspection by the Government. Any critical defects encountered shall be reported within 48 hours. A monthly summary report covering critical defects encountered shall be furnished. The agencies to whom the report shall be furnished are identified in 6.2. The reports shall contain, as a minimum, the following information:

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a. Contractor and contract number.
b. Number and revision date of specification.
c. Lot or serial number.
d. Number of defects found.
e. Assignable cause or analysis of occurrence.
f. Corrective action taken or recommended.

When the corrective action is beyond the control of the contractor or the scope of the contract the report shall contain the contractor's recommendation to the Government.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwg 8786200) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests need not be conducted in order specified in Table II.

Table II

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Operating tolerances (see 3.5)		X
3. Repeatability (see 3.6)		X
4. Response (see 3.7)		X
5. Pneumatic seal (see 3.8)		X
6. Setting mechanisms (see 3.9)		X
7. Resistance (see 3.10)		X
8. Insulation resistance (see 3.11)		X
9. Drift and leakage (see 3.12)		X
10. Low temperature (see 3.13.1)		X
11. High temperature (see 3.13.2)		X
12. Storage (see 3.13.3)		X
13. Temperature shock (see 3.13.4)	X	
14. Vibration (see 3.13.5)	X	
15. Acceleration (see 3.13.6)		X
16. Shock (see 3.13.7)	X	
17. Overpressure (see 3.13.8)		X
18. Under pressure (see 3.13.9)		X
19. Expected life (see 3.13.10)	X	
20. Pressure and contact cycling (see 3.13.11)	X	

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Assembly of M-5 baroswitch prior to assembly of cover, resilient mounts, pneumatic systems and terminal plate assembly, (see dwgs. 8786179, 8803825, 8803840, 8803842, 8803843, and 8803044 covering a detail of dwg. 8786200).

Categories	Characteristics	Method of Inspection
Critical:		
	1. "O" Ring not installed in groove of plate	Visual
	2. "O" Ring missing from tapered hole	Visual
	3. Terminal and plate wire connected to the wrong plate and terminal	Visual/Manual
Major:		
	101. Solder connections from plate and terminal assembly to terminal on diaphragm loose	Visual
	102. Improper solder connection of plate and terminal (8803825) or terminal and plate (8786179)	Visual/Manual
	103. Locating pin in plate assembly loose or bent	Visual/Manual
	104. Soldering improper (see 3.15)	Visual
	105. Front element mounting screws loose	Manual
	106. Wire bare of insulation at terminals	Visual
	107. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.16)	Visual
	202. Hardware loose or missing	Visual/Manual

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4.3.2.1.2 Assembly of M-5 baroswitch prior to assembly of cover and resilient mounts (see dwgs. 8803840, 8803842, 8803843, 8803844, and 8803870 covering a detail of dwg. 8786200).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Chips or other foreign matter in gear teeth	Visual
102.	Evidence of nicks, scratches, burrs or deformation resulting in protrusions, slivers, feather edges, on the gear	Visual
103.	Any gear improperly meshed with its mating gear	Visual
104.	Terminal board free from excess solder or foreign matter	Visual
105.	Any bent or broken pins of connector	Visual
106.	Any evidence of breaks or peeling of connector potting compound	Visual
107.	Gas evacuation tube broken or not sealed	Visual
108.	Protective caps missing from pressure fitting	Visual
109.	Any moving part of gear train assembly in contact with wires	Visual
110.	Torn or split insulating sleeving	Visual
111.	Protective shielding on harness assembly torn or frayed	Visual
112.	Damage to plating exposing base metal	Visual
113.	Terminal plate assembly loose (8786179)	Visual/Manual
114.	Soldering improper (see 3.15)	Visual
115.	Any solder connection loose or poorly connected to a terminal	Visual/Manual
116.	Any spacer loose or bent	Visual/Manual
117.	Any gear improperly secured to its respective shaft	Visual/Manual
118.	Any component part or gear train assembly loose or missing	Visual/Manual
119.	Any wire from servo motor loose or broken	Visual/Manual
120.	Adapter improperly secured to plate	Visual/Manual
121.	Baroswitch elements not oriented or improperly secured	Visual/Manual
122.	Tee fitting loose or missing	Visual/Manual
123.	Any springs loose or missing (see 8803877)	Visual/Manual
124.	Objective evidence relating to material missing or improper (see 3.1)	Visual/Manual

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Minor:

201.	Dents or nicks in servo motor or transmitter	Visual
202.	Marking misleading or unidentifiable	Visual
203.	Evidence of poor workmanship (see 3.16)	Visual
204.	Any flat head screw loose or protruding	Visual/Manual
205.	Serialization missing or improper (see 3.4)	Visual

4.3.2.1.3 Assembly of M5 Baroswitch (see dwg. 8803846 covering a detail of dwg. 8786200).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101.	Protrusion of isolator belt incorrect (0.468 ± 0.031 inch)	Gage
102.	Any isolator incorrectly located	Gage
103.	Any connector pins bent or broken	Visual
104.	Damage to finish exposing base metal	Visual
105.	Rubber gasket torn, twisted, missing or otherwise defective	Visual
106.	Any of three required protective caps missing or cracked	Visual
107.	Soldering improper (see 3.15)	Visual
108.	Any required hardware loose or missing	Visual/Manual
109.	Any required terminal loose or missing (see dwg 8803863)	Visual/Manual
110.	Tee fitting loose or missing	Visual/Manual

Minor:

201.	Evidence of poor workmanship (see 3.16)	Visual
202.	Marking misleading or unidentifiable	Visual
203.	Name plate loose or improperly oriented	Visual/Manual
204.	Serialization improper or missing (see 3.4)	Visual

4.3.2.1.4 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection

Critical: None defined.

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Major:

101. Preservation and packaging not in accordance with MIL-STD-1169.....Visual
102. Wrapping paper not in accordance with MIL-P-17667, Type II, Class I.....Visual
103. Bag torn or not sealed properly.....Visual
104. Bag not in accordance with MIL-B-117, Type I, Class C.....Visual
105. Marking not in accordance with MIL-STD-1169.....Visual

Minor:

201. Evidence of poor workmanship (see 3.16).....Visual

4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified below:

	Test	Test Classification
a.	Insulation resistance	Major
b.	Resistance	Major
c.	Pneumatic seal	Major
d.	Setting mechanisms	Major
e.	Operating tolerance	Major
f.	Repeatability	Major
g.	Response	Major
h.	Drift and leakage	Major
i.	Low temperature	Major
j.	High temperature	Major
k.	Synchro calibration	Major
l.	Overpressure	Major
m.	Under-pressure	Major
n.	Contact chatter	Major
o.	Shock	Major
p.	Pressure and contact cycling	Major
q.	Expected life	Major
r.	Acceleration	Major
s.	Storage	Major

Of these tests, (n) through (q) are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

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4.3.3.1 Pneumatic seal, insulation resistance, resistance, setting mechanisms, operating tolerance, and repeatability, response, drift and leakage, low temperature tests, high temperature tests and synchro-calibration. Each baroswitch assembly shall be tested in accordance with 4.4.3, 4.4.7, 4.4.5, 4.4.6, 4.4.1, 4.4.2, 4.4.8, 4.4.9.1, 4.4.9.2, and 4.4.10. If any item fails to comply with 3.8, 3.11, 3.10, 3.9, 3.5, 3.6, 3.7, 3.12, 3.13.1, 3.13.2 and 3.9.3, the item shall be rejected and removed from the lot.

4.3.3.2 Over-pressure and under-pressure. Sample baroswitches shall be selected in accordance with MIL-STD-105 utilizing as Acceptable Quality Level (AQL) of 2.5 percent, Lot Tolerance Percent Defective (LTPD) of 14 percent with lot and sample size based on the number of bse. All samples selected shall be tested in accordance with 4.4.9.8 and 4.4.9.9. For lots of 300 or less if any item fails to comply with 3.13.8 and 3.13.9 the lot shall be rejected.

4.3.3.3 Contact chatter, shock, pressure and contact cycling, expected life, storage and acceleration. One sample baroswitch assembly shall be selected from each 100 units produced until the first 300 units are produced. Thereafter one sample shall be selected from every 300 units produced. All samples selected shall be tested in accordance with 4.4.9.5, 4.4.9.5.1, 4.4.9.7, 4.4.9.11, 4.4.9.10, 4.4.9.3.1 and 4.4.9.6. If any item fails to comply with 3.13.5.1, 3.13.7, 3.13.11, 3.13.10, 3.13.3 and 3.13.6, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-8786200. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, tests shall be conducted at ambient room temperature.

Acceleration	± 5 percent
Remote indicator setting	± 5 feet
Current	± 5 percent
Altitude	± Pressure equivalent of ± 6.6 mbs. at normal altitude
Temperature	± 5°F
Time	± 2 percent
Vibration amplitude	± 5 percent
Vibration frequency	± 2 percent
Voltage	± 5 percent

4.4.1 Operating tolerance and repeatability. For each bsa, perform three dives (see 6.11.6) at each of the normal altitude settings shown in Table I (see 3.5). Then record the operating point for each bse on each dive. All operating points shall not exceed the tolerances shown in Table I. Repeatability for each bse for the three dives at each setting shall not exceed the limits shown in Table I. Observe and record each reading in mbs.

4.4.2 Response. Establish a nominal altitude setting of 4500 feet on the bsa. Perform three dives starting with the bsa pneumatic system evacuated to 10.11 mb. From this point the rate of change of pressure in the pneumatic system through the operating point shall be 133, (+27, -0) mb/seconds. After completion of this test, the bsa shall be tested as specified in 4.4.1.

4.4.3 Pneumatic seal. The bsa pneumatic system shall be subjected to a pressure 666 + 6.6 mb below external pressure. Using a mercurial manometer connected to the pneumatic system an observation shall be made of a change in pressure in the system over a given period of time. It is suggested that the interval of time of observation not be less than one minute nor more than three minutes. Record rate observed in mb per minute.

4.4.4 Range setting limits. Connect 26 ± 2 volts, 400 Hz ac power and a capacitor (see 6.12) to the bsa setting motor and synchro circuits. Operate the bsa setting system to test for proper operation of the bsa setting range limit switches. Observe and record each reading in feet.

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4.4.5 Remote setting rate. The bsa shall be remotely set to a nominal altitude of 25,000 feet. The setting of the bsa shall then be changed to -3000 feet without any running warmup of the motor. The elapsed time for setting over this range shall be observed and recorded. During the test the electrical input to the setting motor shall be 26 ± 2 volts, 400 ± 10 Hz with a capacitor connected as indicated in 6.12.

4.4.6 Resistance.

4.4.6.1 Continuity loop. Apply a non-inductive current of 100 ma at a potential of 30 volts dc max. between pins X and B of the connector. Observe and record reading in ohms.

4.4.6.2 Element contact loop. With a nominal setting of 4500 feet on the bsa, and using the circuit defined in Figure 1, decrease the pressure to 20.6 ± 6.7 mb less than the nominal operating point of the bsa. Increase the pressure applied to the pneumatic system slowly to permit observation of pressure at instant of contact closure. When contact closure is observed, the pressure shall be further increased 3.3 ± 0.3 mb switch to the circuitry used to measure loop resistance and determine the loop resistance when the contact is conducting 300 ± 20 ma in a 30-volt maximum dc circuit. The above procedure shall be repeated for each bse. Observe and record each reading in ohms.

CAUTION: Do not allow the contacts to make or break the 300-milliampere current.

4.4.6.3 Synchro rotor windings. The dc resistance of the two synchro rotors in parallel shall be measured by a method providing an accuracy of ± 5 percent when compared with measurements made on a standard bridge at 77°F. The dc resistance shall be measured at pins T and J of the connector. Current during this test shall not exceed 300 ma and shall not be applied for more than five seconds. Observe and record reading in ohms.

4.4.7 Insulation resistance.

4.4.7.1 Element circuits.

4.4.7.1.1 Low altitude. With the bsa remotely set to -1000 feet and the pneumatic system subjected to a 1070 ± 5 mb's feet pressure altitude use a megohm bridge or other equivalent instrument and apply a potential of 500 ± 50 volts dc successively, between connector pins C, P, V, M and the case. All unenergized connector pins except D, W, L and N shall be grounded to the case. In the test circuit, protective resistance shall be supplied to limit the current to 200 ± 20 microamperes under short circuit conditions. Observe each reading in megohms.

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4.4.7.1.2 High altitude. With the bsa remotely set to 25,000 feet and the pneumatic system subjected to a pressure of 10 ± 5 mb's use a megohm bridge or other equivalent instrument and apply a potential of 300 ± 30 volts dc successively between connector pins C, D, P, W, V, L, M, N and the case. All unenergized connector pins not receiving this test shall be grounded to the case. In the test circuit protective resistance shall be supplied to limit the current to 120 ± 12 microamperes under short circuit conditions. Observe each reading in megohms.

4.4.7.2 Synchro and motor circuits. Using a megohm bridge or other equivalent instrument apply a potential of 375 ± 25 volts rms, 60 Hz ac between pin T of the connector and the case current to be limited to 120 microamperes. The voltage applied shall be limited to a maximum of five seconds duration. Observe and record each reading in megohms.

4.4.8 Drift and leakage. The bsa shall be tested at room atmospheric conditions for stability of the bse. The purpose of the test is to determine that the bse will not be subject to an excessive shift in contact operating point during storage due to leakage or drift. Leakage is indicated by a lower-altitude closure (plus pressure deviation) than the initial altitude closure. Drift is indicated by a higher altitude closure (minus pressure deviation than the initial altitude closure).

4.4.8.1 Initial operating readings. At a nominal altitude setting of 4,500 feet on the bsa, determine the operating pressure of each bse using the average of three dives through the operating points.

4.4.8.2 Test procedures. After determining the initial operating pressure of each bse, the bsa shall be stored for two or four successive 15 day periods, except where rejection occurs earlier, with operating point tests to be made per procedure of 4.4.8.1 at the end of each 15 day period. With the operating point data obtained at the end of each 15 day period, the deviation (algebraic difference) in operating pressure from the initial operating readings shall be determined for each bse. Observe and record each reading in mb's.

4.4.8.3 Acceptance.

4.4.8.3.1 Thirty (30) day drift period. An acceptable bsa shall meet the performance requirements of a and b in this paragraph. BSA's which do not meet these requirements shall be re-examined with respect to the requirements of a and b in 4.4.8.3.2, and if acceptable for further testing, shall be continued on drift for an additional 30 days (total time 60 days).

a. The positive (+) deviations in operating point of any bse from the initial operating point determined in 4.4.8.1 and the operating points determined at the end of each of the two 15-day periods shall not exceed 1.0 mb.

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b. The negative (-) deviations in operating point of any bse between the initial operating point determined in 4.4.8.1 and the operating points determined at the end of each of the two 15-day periods shall not exceed 1.0 mb.

4.4.8.3.2 Sixty (60) day drift period. An acceptable bse shall meet the performance listed below. The bsa shall be rejected at any time it does not meet the requirements of this paragraph during the course of the 60-day drift test.

a. The positive (+) deviations in operating point of any bse from the initial operating point as determined in 4.4.8.1 and the operating points determined at the end of the four 15-day periods shall not exceed 2.0 mb.

b. The negative (-) deviations in operating point of a bse between the initial operating point determined in 4.4.8.1 and the operating points determined at the end of the first and second 15-day periods shall not exceed 2.6 mb.

c. The negative (-) deviations in operating point of a bse between the initial operating point determined in 4.4.8.1 and the operating points determined at the end of the third and fourth 15-day periods shall not exceed 3.9 mb.

d. The positive differential between the 30th day reading and the 60th day reading shall not exceed 1.0 mb per bse.

4.4.8.4 BSA recalibration. If necessary, after the bsa has been found acceptable per the requirements of 4.4.8.3, the bsa shall be recalibrated to assure optimum operation within the requirements of 3.5 and 3.6.

4.4.9 Environmental testing (see 3.13).

4.4.9.1 Low temperature.

4.4.9.1.1 Remote setting rate. Prior to subjecting the bsa to -65 ± 5°F the bsa shall be remotely set to a nominal altitude of 25,000 feet. After the bsa temperature is stabilized (see 6.4) at -65 ± 5°F, the setting of the bsa shall be changed to -3000 feet without any running warmup of the motor. The elapsed time for setting over this range shall be noted and recorded. During the test the electrical input to the setting motor shall be 26 ± 2 volts, 400 ± 10 Hz with a capacitor connected as indicated in 6.6.

4.4.9.1.2 Pneumatic seal. At a stabilized (see 6.4) temperature -65 ± 5°F the bsa shall be tested at two bsa altitude settings of -3000 and 25,000 feet. The bsa pneumatic system shall be subjected to a pressure of 666 ± 6.7 mb below external pressure. With a total system volume of 10 cubic inches, leakage shall not exceed a rate which will cause a 8 mb per minute rise in pressure in the bsa pneumatic system. For test systems of any other total volumes, the allowable pressure change rate shall be inversely proportional to the volume ratio; for a system of 20 cubic inches (twice the above volume) the allowable leakage rate would be one-half or 4.0 mb/minute.

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4.4.9.1.3 Operating tolerance and repeatability. Stabilize each bsa at -65 ± 5°F and perform three dives (see 6.11.6) at each of the nominal altitude settings shown in Table I (see 3.5). Then record the operating point for each bse on each dive. Observe and record each reading in mb's.

4.4.9.2 High temperature.

4.4.9.2.1 Remote setting rate. Prior to subjecting the bsa to +200 ± 5°F, the bsa shall be remotely set to a nominal altitude of 25,000 feet. After the bsa temperature is stabilized (see 6.4) at +200 ± 5°F., the setting of the bsa shall be changed to -3000 feet without any running warmup of the motor. The elapsed time for setting over this range shall be recorded. During the test the electrical input to the setting motor shall be 26 ± 2 volts, 400 ± 10 Hz with a capacitor connected as indicated in Note 6.6.

4.4.9.2.2 Operating tolerance and repeatability. Stabilize (see 6.4) each bsa at a temperature of 200 ± 5°F and perform three dives (see 6.6) at each of the nominal altitude settings shown in Table I. Then record the operating point for each bse on each dive. Observe and record each reading in mb's.

4.4.9.3 Storage. Electrical and pneumatic fittings shall be capped during these tests.

4.4.9.3.1 Low temperature. Perform three to ten dives at room temperature and a 4500 foot altitude setting before this test. The average operating point shall be determined using data from all dives performed. The bsa shall then be subjected to a temperature of -65 ± 5°F until temperature stabilization (see 6.4) is reached and then to a temperature of -80 ± 5°F for 48 hours. The temperature shall then be raised to -65 ± 5°F and maintained for 13 days. After the low temperature test, from three to ten dives shall be performed at room temperature and a 4500 foot altitude setting. After the low temperature storage, tests 4.4.1, 4.4.3 through 4.4.7 and 4.4.10 shall be performed at room temperature.

4.4.9.3.2 High temperature storage. Perform three to ten dives at room temperature at a 4500 foot altitude setting before this test. The average operation point shall be determined using data from all dives performed. The bsa shall then be placed in a chamber and the following 24 hour cycle performed 30 consecutive times.

Cumulative Time (Hours)	Temperature (°F)
10	90
10-15	Change to 160
15-19	160
19-24	Change to 90

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The relative humidity shall be maintained at 15 percent or less. During the test, a 2-foot ± 1-inch length of 3/8-inch hose shall be attached to the pressure fitting. After this test, from three to ten dives shall be performed at room temperature, and a 4500 foot altitude setting. The average operating point shall be determined using data from all dives performed. After the high temperature storage tests per 4.4.1, 4.4.3 through 4.4.7 and 4.4.10, shall be performed at room temperature.

4.4.9.4 Temperature shock. The assembly shall be tested in accordance with MIL-E-5272, Procedure I, Temperature Shock, except that the high temperature shall be 160 ± 5°F and the low temperature shall be -65 ± 5°F. When operated at the end of the test period the assembly shall be tested as specified in 4.4.1, 4.4.3 through 4.4.7 and 4.4.10.

4.4.9.5 Vibration. The switch, with isolators, shall be mounted on a suitable vibration table. Provisions shall be made for vibrating along each of the three major mutually perpendicular axes of the switch. One axis shall be parallel to the longitudinal axis of the vibration isolator mounting bolts. Another axis shall be perpendicular to the surface of the bsa case mounting and pressure fitting. The third axis shall be perpendicular to both of those defined above. The monitoring circuit and instruments used shall conform to the following requirements:

a. A closed bse contact shall conduct less than 50 microamperes.
b. The voltage across the contact shall be dc and shall at no time exceed 30 volts.
c. A single make of a normally open contact, or break of a normally closed contact, of 50 microseconds duration shall always give a definite indication.

4.4.9.5.1 Test. The bsa shall be remotely set to a 4500 foot altitude setting. Three dives shall be performed to establish the average operating pressure of each bse. The chatter monitoring device shall be connected across the bse contact circuits. Subject the bsa cavity to an absolute pressure 8 mb less than the average operating pressure of the bse having the lowest operating pressure. Subject the bsa vibration inputs as specified below, and monitor the bse contacts for contact chatter. Subject the bsa cavity to an absolute pressure 8 mb greater than the average operating pressure of the bse having the highest operating pressure and repeat the test. The above test shall be performed along each of the three mutually perpendicular axes.

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Frequency Range (Hz)	Constant Acceleration (g) or Double Amplitude (inch)	Duration of Scanning
15-73	0.036 inch	2 to 5 minutes
73-500	10.0 g	5 to 10 minutes

During the vibration there will be no indication of contact chatter. After this test, the bsa shall be tested as specified in 4.4.1, 4.4.3, 4.4.4, 4.4.6 and 4.4.7.

4.4.9.5.2 Resonance.

4.4.9.5.2.1 Search. Resonant frequencies shall be determined by vibrating the assembly in each of three major mutually perpendicular axes at room temperature. The frequency of the applied vibration shall be varied slowly from 15 to 500 to 15 Hz at low amplitudes. Amplitudes shall not exceed those specified in 4.4.9.5.1. The tests shall be made in such a manner as to detect major resonances of any isolation system, internal or external, or of the primary sensing element.

4.4.9.5.2.2 Vibration at resonance. The assembly shall be vibrated at each of the significant resonant conditions observed in 4.4.9.5.2.1 for one hour at room temperature, 15 minutes at -65°F., and 15 minutes at 160°F. If more than one resonant condition was encountered in the same axis, the test period may cover the condition which is considered the most likely to produce failure. If no well defined resonances were apparent the assembly shall be vibrated for the periods specified above at the condition or conditions considered most likely to produce failures. Following this test, the resilient mounts may be replaced. The amplitudes of vibration shall conform to those listed below.

Frequency Range (Hz)	Constant Acceleration (g) of Double Amplitude (inch)
15-73	0.036 inch
73-500	10.0 g

After this test, the bsa shall be tested as specified in 4.4.1, 4.4.3 through 4.4.7 and 4.4.10.

4.4.9.5.2.3 Cycling. The assembly shall be vibrated at an applied double amplitude of 0.060 inch through the frequency range of 10 to 55 Hz in one minute cycles. If resonance search has indicated resonances between 55 and 500 Hz, the range and cycling time shall be extended to include resonances, and the input shall be 10 g acceleration. Cycling shall be performed in each of the three major mutually perpendicular axes for one hour at room temperature, 15 minutes at -65°F., and 15 minutes at 160°F. Following this test, the resilient mounts may be replaced.

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4.4.9.6 Acceleration. The bsa shall be remotely set to a 4500 foot altitude setting. Three dives shall be performed to establish the average operating pressure of each bse. Subject the assembly to 50 g linear acceleration for one minute; then reduce the acceleration to 15 g. During the 15 g acceleration subject the pneumatic system to three dives (see 6.5.6) and determine operating point for each bse. Repeat this test in both directions of three mutually perpendicular axes.

4.4.9.7 Shock. Before the test the bsa shall be set at 4500 feet. The bsa shall then be bolted to a flat plate by means of its isolation assembly. The bsa shall be subjected to 18 impact shocks, 12 of 30 g and 6 of 50 g (g is input value). Two 30 g and one 50 g shock shall be applied in each direction along each of the three mutually perpendicular major axes of the bsa. Each 30 g shock impulse shall reach its maximum within 0.0055 ± 0.001 second and shall have a total duration of 0.011 ± 0.001 second, and each 50 g shock impulse shall reach its maximum within 0.0025 second and shall have a total duration of 0.005 ± 0.001 second.

4.4.9.8 Overpressure. The bsa shall be remotely set to 4500 foot altitude setting. From three to ten dives shall be performed to establish the average of all dives performed. Subject the bsa cavity to a pressure of 1520 ± 5 mb for 20 hours. Immediately at the end of the overpressure period, again subject the bsa to three to ten dives and establish the average operating pressure of each bse.

4.4.9.9 Under-pressure. The bsa shall be remotely set to a 4500 foot altitude setting. From three to ten dives shall be performed to establish the average operating pressure of each bse using the average of all dives performed. Subject the bsa cavity to a pressure of 10.6 mb's or less for 20 hours. Immediately at the and of the under-pressure period, again subject the bsa to three to ten dives and establish the average operating pressure of each bse.

4.4.9.10 Expected life. Apply 26 ± 2 volts 400 ± 10 Hz ac power to the setting motor and synchro circuits of the bsa with a capacitor connected as specified in Note 6.6 and cycle between -3000 to 25,000 to -3000 feet for 300 cycles. During cycling, the motor current shall be interrupted at the ends of the setting range by the bsa limit switches, and a period of time not to exceed one minute may elapse before reversing the direction of drive. On completion of 300 cycles, the bsa shall be tested as specified in 4.4.1, 4.4.3 through 4.4.7 and 4.4.10 except that 4.4.6.1 shall be omitted.

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4.4.9.11 Pressure and contact cycling. Before this test the bsa shall be set at an altitude setting of 4500 feet. Then the bsa pressure cavity shall be subjected to 1000 cycles of pressure change from 1014 to 10.7 to 1014 mb. During the final 500 pressure cycles the bse contacts shall make and break a 100 ± 10 ma current in a non-inductive circuit. Circuit voltage shall be 30 ± 5 volts dc. After this test, the bsa shall be tested as specified in 4.4.1, 4.4.3 through 4.4.7 and 4.4.10.

4.4.10 Synchro-calibration. Using the coarse synchro transmitter as a reference, the deviation of the fine synchro from electrical zero shall be determined by performing a test for calibration between the coarse and fine synchro transmitters at electrical zero (00.000 altitude setting). The test device shall be calibrated against a zero-set transformer or synchro standard to within plus or minus 12 minutes or its equivalent in feet of altitude.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The bsa shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The bsa shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 The bsa shall be wrapped in paper conforming to MIL-P-17667, Type II, Class I and sealed in a bag conforming to MIL-B-117, Type I, Class C.

5.2.2 Levels A and C. The packed bsa shall then be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 and 3.14 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

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6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Definitions.

6.5.1 Baroswitch assembly (bsa). The multiple assembly of the four switch elements on the base plate, the combined setting and remote indicating system, and the external isolation system shall be known as a Baroswitch Assembly.

6.5.2 Baroswitch element (bse). Any one of the four pressure sensing mechanisms shall be known as a Baroswitch Element.

6.5.3 Remote setting indicator. The remote setting indicator is the altitude indicating device connected to the bsa by an electrical connection only, which gives an indication of the nominal altitude setting of the bsa.

6.5.4 Atlitude-pressure relationship. Wherever an altitude or altitude change is specified, it shall be simulated by a pressure or pressure change which conforms to the relationship specified by NACA report No. 1235.

6.5.5 Pneumatic system. The pressure port and the connected passage ways and cavities within the bsa that supply the sensing air to the bse shall be known as the pneumatic system.

6.5.6. Dive. A dive is a change in pressure experienced by the bsa pneumatic system for 50,000 feet through the operating point. The rate of change of pressure from 50,000 feet shall exceed 1000 feet/second to within 1000 feet or 100 mb's of the expected operating altitude of the bsa. Then the rate of change in pressure may be slowed to a rate compatible with test equipment and/or operating personnel; however, this rate shall be not less than 200 feet/minute.

6.5.7 Operating point. The altitude at which the bse contacts close on a dive shall be known as an operating point.

6.5.8 Nominal altitude setting. The setting shown by the remote setting indicator shall be the nominal altitude setting for the bsa.

6.5.9 Operating error. The difference between the altitude at which the bse contacts close on a dive, and the nominal altitude setting for the bsa shall be known as the operating error.

6.5.10 Repeatability. The maximum difference between operating point for any one bse for two or more dives during any 24-hour period when pressure is the only variable (no switch adjustments or changes in temperature) shall be known as repeatability.

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6.5.11 Loop resistance. The resistance of a closed contact circuit contained within the bsa, measured from connector pin to connector pin, shall be known as loop resistance.

6.5.12 Room temperature. The room temperature shall be 80 ± 15°F.

6.6 Setting motor capacitor. Operation of the baroswitch setting motor requires that a capacitor shall be connected across pins F and U of the electrical connector external to the bsa. The capacitor used shall have a rating of 2.6 ± 5 percent microfarads at a working 200 volts dc.

6.7 The altitude indicating transmitter system. The altitude indicating transmitter system of the baroswitch assembly incorporates dual synchro transmitters. One shaft rotation of one synchro (the fine) is equivalent to 3600 feet of altitude. The second synchro (the coarse) is geared down to a 20 to 1 ratio with the fine synchro so that one shaft rotation is equivalent to 72,000 feet of altitude.

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6.8 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1043 (Rev. 6) 13 July 1964.

Custodian:
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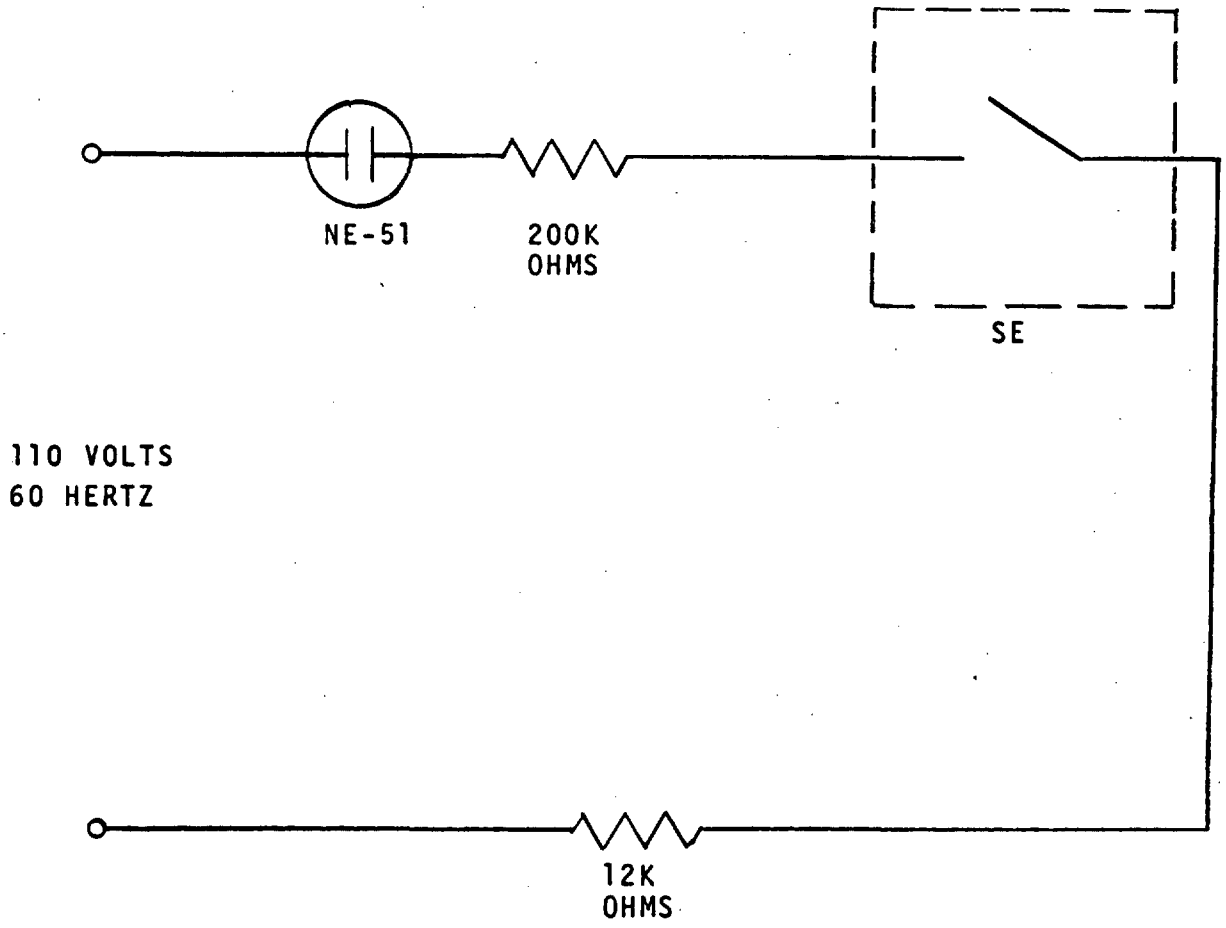


Figure 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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